



Summer Undergraduate Research Award

Stochastic Reaction Network Modeling and Optimal Control of Bacterial Populations Under Antibiotic Stress

Investigating Antibiotic Resistance Emergence under Entropic Control Theory

By,

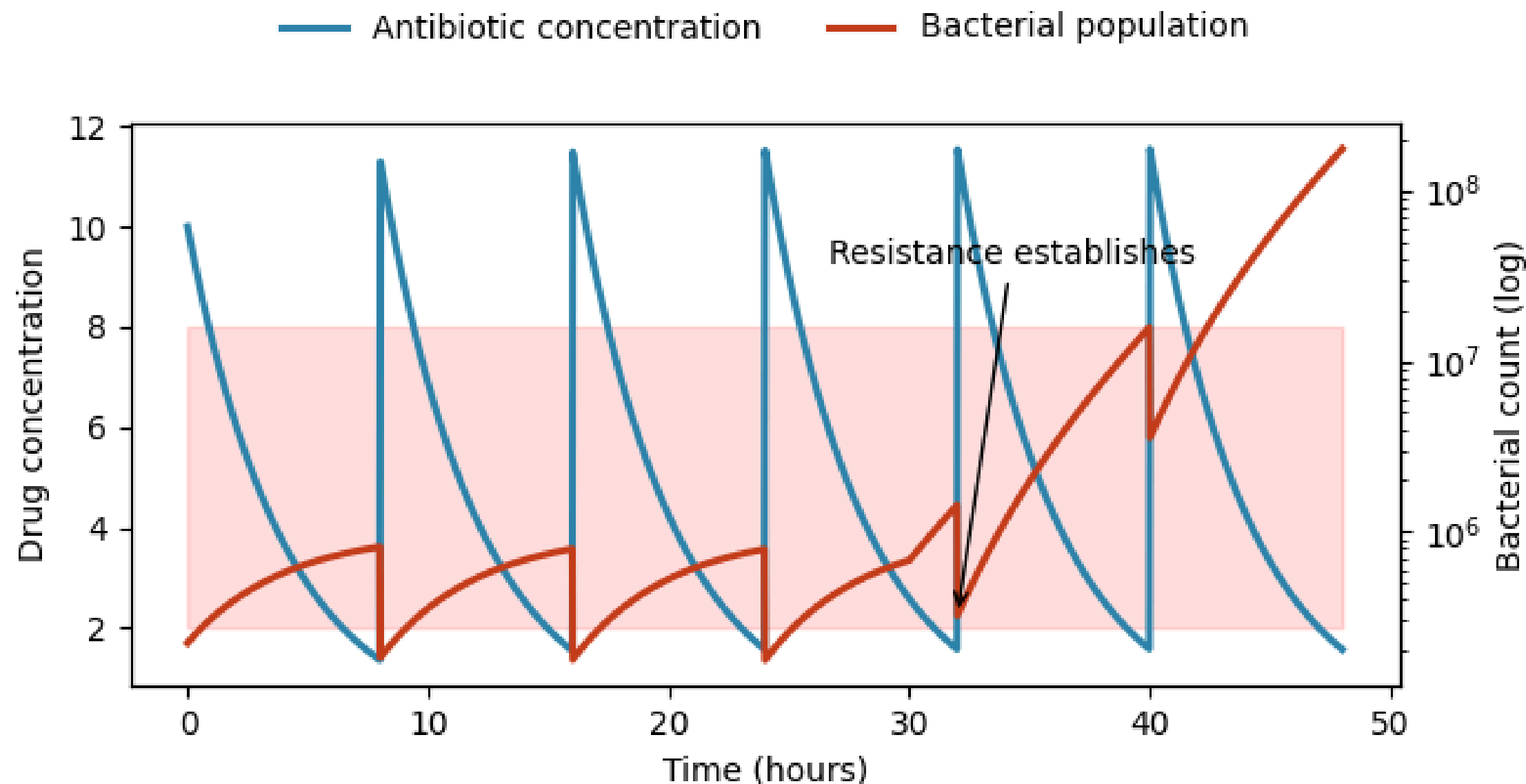
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A Silent Pandemic: Antibiotic Resistance

- **~1.27 million** deaths attributed to antimicrobial resistance annually (Lancet 2022)
- Projected 10 million/year by 2050 if unchecked
- Fixed dosing schedules are blind to stochastic resistance emergence



Why current protocols are suboptimal?

- Fixed schedules ignore pharmacokinetics
- Average-case design misses tail risks
- No state-dependent feedback

Where the Theory Stands — and Where It Stops

Stochastic Reaction Networks (SRNs)

Model populations as discrete counts changing via random reaction events — the natural language for small, noisy biological systems.

Horiguchi & Kobayashi (PRX Life, 2025)

- Formulated optimal control for SRNs using KL-divergence cost
- Linearizes the Hamilton-Jacobi-Bellman equation → tractable solutions
- Discovered mode-switching: controller alternates ON/OFF phases

Critical Gaps

- Rates assumed constant in time (real drugs decay between doses)
- No mutation reaction (so resistance can't even emerge in their model)
- Outcomes measured by averages, not by rare-event probabilities
- Computationally infeasible for large populations — direct HJB solution doesn't scale

How the Theory Actually Works??

The Horiguchi-Kobayashi Algorithm

STEP 1 – Define Reactions & Propensities

- Reactions with stoichiometry
- Propensity function gives reaction likelihood at state n
- Rate coefficients k_r are our control variables

$$r \in R, \quad s_r \in \mathbb{Z}^{|X|}, \quad h_r(n) \geq 0, \quad k_r \geq 0$$

STEP 2 – Choose the Cost Function

- Measure control effort as KL divergence
- Compares controlled dynamics vs natural dynamics

$$C_{KL}(n, k) = \sum_{r \in R} h_r(n) \left[k_r \log \frac{k_r}{k_r^0} - k_r + k_r^0 \right]$$

STEP 5 – Compute Optimal Rates

$$k_r^\dagger(t, n, \beta) = k_r^0 \exp \left[\nabla_{s_r} V_t(n, \beta) \right]$$

- Exponential enforces non-negativity
- Depends on state n and time t

STEP 3 – Write the HJB Equation

- Dynamic programming gives value function $V_t(n, \beta)$

$$-\frac{\partial V_t(n, \beta)}{\partial t} = \beta U_t(n) + \max_{k \geq 0} \left[\mathcal{L}_k V_t(n, \beta) - C(n, k) \right]$$

- Nonlinear · Infinite-dimensional · Intractable

STEP 4 – Linearize via Cole-Hopf ★

- Substitute $Z_t = \exp(V_t)$

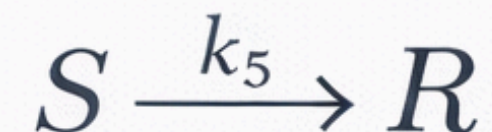
$$Z_t(n, \beta) := \exp[V_t(n, \beta)]$$

$$-\frac{\partial Z_t(n, \beta)}{\partial t} = \beta U_t(n) Z_t(n, \beta) + \mathcal{L}_{k^0} Z_t(n, \beta)$$

- Equation becomes linear → tractable
- Solving the linear equation exactly requires storing one value per state – feasible only for small N .

From Abstract Theory to Bacterial Reality

A Two-Species Stochastic Reaction Network



Rates

↑ Sensitive growth

💊 $k_2(t)$ Drug-induced killing — time-varying

↑ k_3 Resistant growth

💀 k_4 Resistant natural death

🧬 k_5 Mutation: $S \rightarrow R$

The two bolded reactions are absent from Horiguchi & Kobayashi (2025).

What's New: "From Paper → Our Extension"

We extend their framework from abstract biology to clinical antibiotic dosing.

Horiguchi & Kobayashi (2025)

This Project

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Constant reaction rates
.....

Time-varying $k_2(t)$ following pharmacokinetics
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Abstract populations (Moran, SIR)
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Sensitive + Resistant bacteria with explicit mutation
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Average outcomes
.....

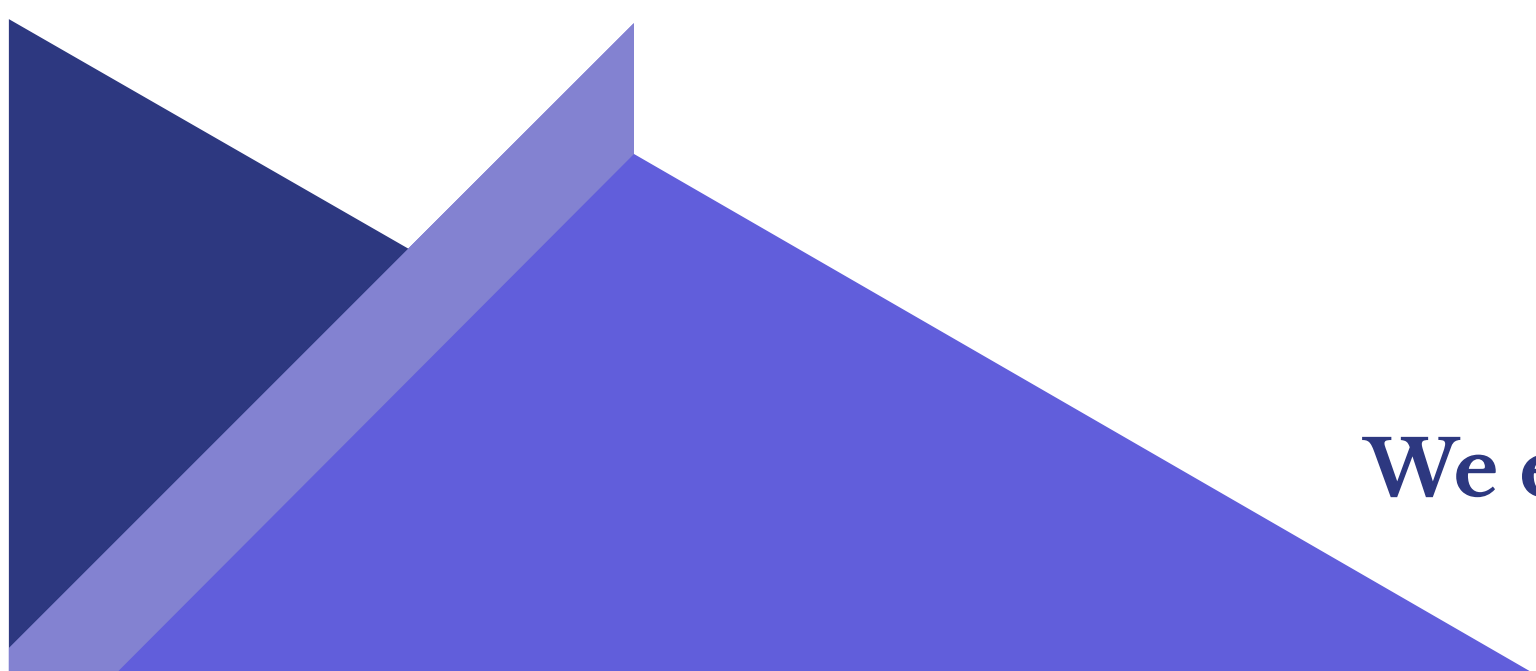
Rare-event probabilities of resistance
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.....
Small N, exact master equation
.....

Scalable Monte Carlo for large N
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.....
Fixed KL-control framework
.....

Extension + speculative actor-critic RL
.....



We extend their framework from abstract biology to clinical antibiotic dosing.

Milestones that Matter

Deliverables

Adaptive Dosing Framework

computational tool for antibiotic schedules under pharmacokinetic constraints

Mode-Switching Schedule Discovery

identify ON/OFF dosing patterns in the bacterial setting

Resistance Emergence Quantification

numerical estimates of rare-event probabilities

Scalable Monte Carlo Simulation

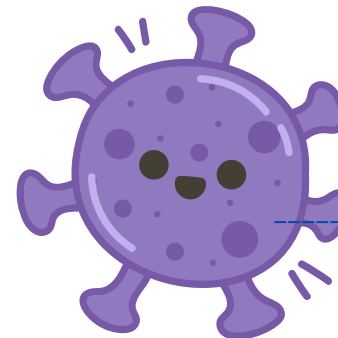
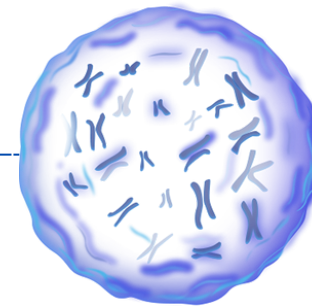
methodology for clinically-relevant population sizes

Why It Matters

Scientific & Clinical Impact

Immediate (Scientific)

- **Extends Horiguchi & Kobayashi's mathematical theory into a clinically-relevant biological domain**
- **Quantitative handle on a rare event that currently has none**
- **a theoretical basis for why "pulse-then-rest" dosing schedules work empirically**

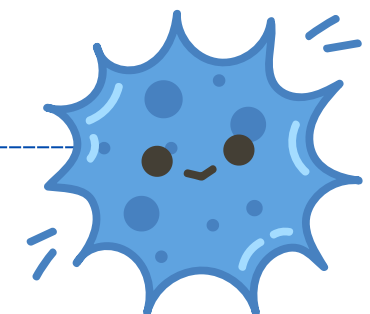


Medium-term (Translational)

- **Adaptive dosing protocols** — designed around each drug's pharmacokinetic profile, not a one-size-fits-all schedule
- **Decision-support tool for clinicians** — estimate resistance risk before prescribing a regimen
- **Benchmark platform** — any future resistance model (multi-drug, multi-strain) plugs into the same framework

Long-term (Societal)

- **Contribution to the AMR fight** — antimicrobial resistance causes 1.27M deaths/year; even a marginal improvement in protocol design has enormous population-scale benefit
- **Cross-domain reusability** — the same mathematical framework applies to cancer-drug resistance, pest resistance in agriculture, and evolutionary rescue in conservation biology



Budget & Timeline

Item	Amount (₹)			
SSD / External Storage	12000		1-2	• Implement Gillespie SSA from scratch • Reproduce paper's benchmark results (Moran, SIR) • Validation checkpoint before extending
RAM Upgrade	7000			
Research Tools & Literature.	10000		3-4	• Construct S/R reaction network • Inject pharmacokinetic $k_2(t)$ profile • Measure baseline resistance emergence rates
Cloud / HPC Credits	2000			
Books & Documentation	3000			
Contingency	6000		5-6	• Extend HJB for time-varying rates • Rare-event Monte Carlo (importance sampling) • Build cost-vs-safety Pareto frontiers
Extra Scientific Software	10000			
TOTAL	50000		7-8	• Technical report with figures • Parameter inference from trajectory data • Actor-critic RL (speculative)

The image features a decorative graphic in the top-left corner consisting of several overlapping triangular and quadrilateral shapes in various shades of purple, ranging from a deep indigo to a light lavender. These shapes create a layered, geometric effect. The rest of the image is a solid, clean white background.

Thank You